Full Stack Assignment – 02

Assignment ID: FSD-02
Title: Slot Booking Management System
Duration: 3 Hours

1️⃣ Objective

Design and develop a **Slot Booking Management System** that enables users to book time-based slots while allowing managers to control availability.
The system must prevent scheduling conflicts and ensure **data persistence using a database**.

2️⃣ User Roles

USER

- Register and log in to the system
- View available slots
- Book a slot
- View own booking history

MANAGER

- Log in to the system
- Create and manage slots
- View all bookings
- Block or unblock slots

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** at API and UI levels

Slot Management

- MANAGER can create time-based slots
- Slots must be stored in the database
- Slot availability must be fetched dynamically

Slot Booking

- USER can book an available slot

- Booking details must be stored in the database
- USER should be able to view booking status

4️⃣ Business Rule (Critical)

⚠️ Overlapping time slots are not allowed.

A slot must not overlap with any existing slot or booking for the same resource/time range.

5️⃣ Technical Constraints (Mandatory)

- All data **must persist in a database**
- Hardcoded or in-memory data storage is **not allowed**
- Slot availability and bookings must reflect real database state
- JWT authentication and role validation are mandatory
- Sensitive configuration must use environment variables

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and MANAGER
- Slot listing view with availability status
- Booking action for USER
- Slot management actions for MANAGER

UI aesthetics are not evaluated; correctness and clarity matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview

- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links